

Received 29 October 2025, accepted 17 November 2025, date of publication 25 November 2025,
date of current version 4 December 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3636732

RESEARCH ARTICLE

Multi-Resolution Data Fusion for Resilient Flood Mapping

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This work was supported in part by the National Oceanic and Atmospheric Administration (NOAA) under Award NA21OAR4590358, in part by the National Aeronautics and Space Administration (NASA) under Award 80NSSC23M0051, and in part by NSF under Grant 2401942.

ABSTRACT Floods are among the most destructive and costly natural disasters worldwide, necessitating accurate and timely mapping for emergency response and impact assessment. Remote sensing offers a valuable means to monitor inundation, yet trade-offs in spatial, spectral, and temporal resolution limit single-sensor approaches. This study presents a comparative analysis of flood mapping using multimodal sensor fusion and semantic segmentation. We integrate high-resolution UAV RGB imagery with Synthetic Aperture Radar (SAR) data from Sentinel-1 and UAVSAR, focusing on the classification of four land cover classes relevant to flooding: inundated vegetation, dry vegetation, open water, and others. A modified U-Net architecture was developed to handle multimodal stacked inputs, allowing systematic evaluation of different sensor combinations. The results show that the combination of RGB UAV with UAVSAR VV polarization at 5 m resolution achieves the highest performance (mean IoU: 84.9%), outperforming RGB UAV alone (77.8%) and significantly surpassing the combinations of higher resolution UAVSAR (1 m, 61.2%) and Sentinel-1. These findings highlight the complementary strengths of optical and SAR features, the importance of contextual information, and the counterintuitive benefit of medium-resolution UAVSAR for scene-level understanding. Incorporation of SAR indices such as VH/VV ratios provided modest gains, particularly in vegetation-rich areas. Cross-domain testing with Hurricane Matthew imagery revealed a notable drop in generalization (best mean IoU: 46.2%), underscoring challenges of domain shift and the need for domain-adaptive modeling. Overall, the results emphasize the advantages of multimodal fusion, especially UAV RGB with UAVSAR VV (5 m), for fine-grained flood mapping and suggest future directions in multi-temporal and adaptive frameworks for robust flood monitoring.

INDEX TERMS Semantic segmentation, flood mapping, multimodal data fusion, UAV imagery, synthetic aperture radar (SAR).

I. INTRODUCTION

In recent years, the combined effects of climate change and human-induced environmental changes have led to a rise in the frequency and intensity of natural disasters such as droughts, wildfire, and floods [1], [2], [3]. Among these natural disasters, floods have emerged as the most devastating and economically impactful, leading to extensive harm to infrastructure, property, and human lives on a global scale [4], [5], [6]. Thus, accurate and timely flood mapping

is essential for effective emergency response planning and damage assessments during and after flood events [7], [8], [9]. Remote sensing has been widely used to extract information about the Earth's surface and evaluate the impact caused by floods and planned emergency responses in multiple studies. The increasing availability of diverse remote sensing data from various satellite, airborne, and ground-based sensors has opened up new possibilities for improving flood mapping and damage assessment [10], [11], [12]. In flooded areas, the ability to acquire information remotely proves crucial, especially in extreme situations where ground access might be restricted or dangerous [13]. Researchers have explored

The associate editor coordinating the review of this manuscript and approving it for publication was Donato Amitrano^{ID}.

diverse technologies, including optical, infrared, radar, and Light Detection Radar (LiDAR) sensors to enhance the accuracy and depth of the flood and its damage. One promising approach is the use of multi-resolution sensor fusion, which combines data from sensors with different spatial, temporal, and spectral resolutions to enhance the accuracy and timeliness of flood mapping [14], [15], [16], [17], [18]. This enables the generation of high-resolution flood maps that capture complex urban environments and rapidly evolving flood dynamics [19], [20]. Multi-resolution sensor fusion leverages the complementary strengths of different sensor types, such as the high spatial resolution of optical imagery and the all-weather capabilities of Synthetic Aperture Radar (SAR) data [21], [22]. However, this approach involves a trade-off between spatial, spectral, and temporal resolutions. Reference [23] categorized the trade-offs into spatio-spectral fusion to merge fine spatial and fine spectral resolutions, and spatio-temporal fusion to blend fine spatial and temporal resolutions. Optical satellite images have been used for geohazard studies at regional and local scales, general natural disaster assessment, including flood, landslides, and damage assessment [24], [25], [26], [27], [28], [29], [30], [31], [32]. Optical data have proven to provide results on the degree and extent of damage after flood, and other disasters in general. However, there are several limitations associated with using optical sensors for natural disaster assessment, especially the sensitivity of the sensor to environmental conditions [33]. Factors such as moisture, rain, dust, and cloud cover can affect data acquisition and result in shadow products that hinder accurate mapping efforts [34], [35], [36]. In the past few years, SAR data generated by radar sensors based on microwave signals, have emerged to complement optical sensors by offering all-weather capability and insights into subsurface features that are not visible in optical imagery [37]. Recent research works have used SAR data for monitoring movements and deformations, especially during and after natural disasters such as earthquakes, volcanic activity, floods, and landslides [35], [38], [39], [40], [41]. The availability of high-resolution SAR images in the evaluation of the impact of natural disasters, especially in large areas, is important. However, just as with optical images, there are limitations due to the inherent characteristics of each sensor. SAR imagery suffers from shadowing and layover effects, especially in complex urban environments or areas with dense vegetation which can obscure flood mapping and damage assessment. Moreover, SAR's temporal resolution affects its sufficiency to capture changes in flood dynamics, potentially limiting its effectiveness in real-time flood mapping and damage assessment [42]. In resolving the limitations of SAR data, aerial imagery/Uncrewed Aerial Vehicle (UAV) images address the constraints of using satellite images. UAV images have higher resolution as compared to satellite imagery, resulting in better flood mapping, risk, and damage assessment of the captured area [43]. UAVs can be deployed rapidly and flexibly, allowing for targeted imaging of specific

areas of interest, supplementing SAR data and providing timely updates for dynamic situations and rescue efforts. Thus, it is possible to evaluate which structures are affected by flood events and the extent of damage [44]. In recent years, UAVs have been utilized in research to provide timely information during natural disasters. For instance, identifying buildings and roads submerged in water during flood disasters [45], [46], [47], [48].

Data fusion techniques combine data from multiple sensors and related information from associated databases to achieve improved accuracy and more specific inferences that could not be achieved by the use of a single sensor alone [49]. In the past, numerous methods have been proposed for image fusion [50], and these approaches are primarily performed at three representative levels: pixel-level, feature-level, and decision-level. Pixel-level includes the integration of Panchromatic (PAN) and Multispectral (MS) images acquired from the same sensor, commonly performed to improve image quality. Pixel fusion may cause spatial distortion in the merged image [51]. At the feature level, on the other hand, involves detecting, segmenting, and extracting distinctive features of objects from different domains before merging them [52]. The core of image fusion at the feature level is the accurate selection and extraction of features from multiple images [53]. Improper features could complicate the highlighting of characteristics and may result in incorrect decisions. The decision level then operates at a higher processing stage by integrating extracted features with additional decision rules to strengthen shared interpretations and clarify the observed targets [50]. Features are extracted first from images, and the decision rules are established for these images before merging them. The decision-level fusion is highly dependent on the quality of extracted features, and therefore, results are sensitive to the applied decision rules because they determine their type of features used in fusion processes.

The fusion of remotely sensed images with other information supplied by various ancillary sources has been a key strategy for accurately interpreting complex phenomena such as flooding. The number of fully automated techniques for flood detection using data fusion approaches is increasing. However, to the best of our knowledge, only a few operational flood services leveraging remotely sensed imagery and ancillary data fusion are currently available. For instance, [14] introduced DAFNE, a Matlab-based toolbox for Bayesian multi-source remote sensing fusion and flooding mapping. The results showed the successful fusion of Synthetic Aperture Radar data and optical data to produce detailed probabilistic flood maps that follow the spatial and temporal evolution of inundation. Reference [54] also introduced an unsupervised method for flood detection in urban areas by synergistically fusing SAR intensity and interferometric coherence under the Bayesian network fusion framework. The flood maps produced by the fusion of intensity and coherence alone are validated and produce

an overall accuracy of 94.5%. Reference [55] proposed a framework that fuses multi-spectral imagery from Landsat Analysis-Ready Data (ARD), dual-polarized (SAR), and coastal Digital Elevation Models (DEMs) to produce flood maps at moderate (30 m) spatial resolution. With an overall accuracy of 97%, the study stipulated that higher accuracy is achieved when ARD, SAR, and DEM datasets are readily available and fused.

The integration of multi-modal deep learning has gained increasing attention in flood mapping, particularly through the fusion of UAV imagery, SAR data, and resolution-aware strategies. Studies have shown that combining heterogeneous sensors within deep learning frameworks provides more reliable inundation detection than relying on single-source data. References [56] and [57] introduced a multi-resolution fusion framework that integrates UAV optical imagery with Sentinel-1 SAR, improving flood delineation in complex environments. Likewise, [58] proposed a collaborative approach that fuses SAR, multispectral, and hyperspectral imagery at multiple resolutions, underscoring the critical role of resolution harmonization in coastal flood and wetland monitoring. Other investigations have highlighted UAVSAR fusion for near-real-time flood response, noting that sensor resolution selection substantially influences the accuracy of inundation boundaries. [59], [60], [61]. Collectively, these developments illustrate the growing momentum in multi-modal fusion research for flood mapping and provide context for the novelty of our study, which systematically evaluates polarimetric UAVSAR at different spatial resolutions together with UAV imagery for cross-event flood mapping in North Carolina.

Parallel to these multi-modal fusion efforts, advances in deep learning have also introduced transformer-based and domain adaptation approaches that extend beyond traditional fusion. Transformer architectures, originally designed for natural language processing, have been successfully adapted for multi-modal and remote sensing tasks by capturing long-range dependencies and cross-modal correlations. For instance, hybrid CNN-Transformer frameworks such as SoftFormer [62] (for SAR-optical) fusion and TGF-Net [63] (transformer + gist CNN fusion) have demonstrated how attention mechanisms can fuse optical and SAR features for improved land cover and segmentation accuracy. Parallel to this, domain adaptation methods have been increasingly applied to reduce performance drops when transferring models across events or regions. Approaches such as High-DAN [64] integrate adversarial learning and a high-to-low resolution fusion strategy to improve generalization across cities, while curriculum-style local-to-global adaptation to address patch-level and global domain gaps in remote sensing segmentation. For transformer-based UDA, DAM-Former [65] fuses high-resolution multimodal data with multitemporal information in a transformer framework for unsupervised domain adaptation segmentation. Finally, recent studies on multi-resolution SAR fusion have proposed collaborative strategies to align data across scales: for example, STSNet

[66] addresses cross-spatial-resolution fusion for land cover segmentation, and inter-Sensor High Resolution+ Multi-Temporal fusion [65] integrates high-resolution multimodal and multi-temporal inputs for UDA. These works underscore the importance of resolution harmonization in multi-modal fusion tasks; only a few explicitly tackle flood inundation detection using high-resolution polarimetric SAR data. Our study builds on these advances by systematically evaluating how multi-modal fusion at different spatial resolutions, particularly UAVSAR and UAV imagery, impacts flood mapping accuracy in multiple hurricane events.

In remote sensing, combining data from the same sensor operating in different modes or from various sensors can provide complementary insights. This makes fusing outputs from multiple sensors highly valuable for enhancing the overall performance of remote sensing images, which are essential for human visual interpretation and image processing applications. While UAV/aerial imagery complements satellite imagery with high-resolution spatial details and flexible data acquisition, its weather dependency and limited coverage present challenges. To mitigate these issues, researchers have increasingly focused on integrating UAV/aerial imagery and other satellite data for environmental monitoring. This study aims to contribute to this growing body of work by utilizing diverse remote sensing datasets, including aerial imagery and both satellite and UAVSAR data, to achieve accurate flood mapping in North Carolina.

Overall, existing multimodal fusion studies have demonstrated the potential of combining UAV and SAR imagery for flood mapping; however, systematic evaluation across spatial resolutions, polarizations, and flood events remains limited. The present work addresses this gap by establishing a unified multi-resolution fusion framework that integrates UAV RGB, Sentinel-1, and UAVSAR data to enhance segmentation accuracy and generalization in flood scenarios. The main contributions are summarized as follows:

- A dual-branch U-Net architecture is introduced for multimodal semantic segmentation, enabling joint feature extraction from optical and SAR inputs while preserving modality-specific representations.
- A comprehensive cross-sensor and polarization analysis is performed, examining multiple configurations (VV, VH, HH) and resolutions (1 m and 5 m) to determine their relative effectiveness for flood delineation.
- Resolution–context trade-offs are quantitatively assessed, showing that medium-resolution data can improve robustness by reducing overfitting and enhancing spatial coherence in complex floodplain environments.
- Cross-event generalization is investigated through training on Hurricane Florence and testing on Hurricane Matthew datasets, exposing the influence of domain shift and emphasizing the need for adaptive learning strategies in operational flood mapping.
- An extended version of the DeepFlood dataset is provided, incorporating multi-resolution UAVSAR imagery and spectral characterization to support reproducible

research in multimodal data fusion and resilient flood monitoring.

II. METHODOLOGY

The central objective of this study is to gain insight into the impact of spatial resolutions of remote sensing data for achieving accurate segmentation results. To this end, we developed a comprehensive testing framework to systematically compare the performance of various combinations of input data, resolutions, and models. By testing different sensor combinations and resolutions, our study aims to identify the best inputs and configurations for precise flood mapping. This helps uncover the most effective strategies for delineating inundated vegetation, dry vegetation, open water, and other land cover classes. The research workflow consists of three stages: data preprocessing, fusion and segmentation, and post-processing. The data utilized in this study includes UAV RGB imagery and SAR components from Sentinel-1 (VH and VV polarizations) and UAVSAR (VH, VV, and HH polarizations), analyzed at two spatial resolutions: 1 meter and 5 meters. The following sections describe the processes involved in data pre-processing, the design and implementation of the testing framework, segmentation methods, and post-processing techniques that enhance the precision and utility of the flood mapping outcomes.

A. STUDY AREA AND DATA

Two study areas in the coastal plain region of eastern North Carolina were selected for the research, including Bladen County and the town of Grifton. Both counties are densely vegetated and flood-prone locations that have experienced numerous hurricanes, resulting in substantial rainfall and widespread flooding. In particular, Hurricane Matthew in October 2016 and Hurricane Florence in September 2018 led to catastrophic inundation, with the rural floodplains of North Carolina accounting for 77% of the 53 total fatalities reported during these events [22]. The study areas are illustrated in Figure 2, which shows the sources of UAV, UAVSAR, and Sentinel-1 imagery acquired during the study periods.

A portion of the dataset used in this study is derived from our previously published *DeepFlood* dataset, which was specifically developed for flood mapping applications and includes curated UAV and Sentinel-1 imagery covering the same study regions and hurricane events [56]. This dataset provided a valuable foundation for the current research by enabling the integration of multi-resolution and multi-sensor data sources in a flood-prone geographic context.

The National Oceanic and Atmospheric Administration (NOAA) National Security and Emergency Response provided UAV imagery captured during Hurricane Matthew and Hurricane Florence. These images cover the affected areas in Grifton and Bladen County, North Carolina [67]. UAV photographs were captured at an altitude of approximately 200 feet, providing a very high spatial resolution of 25 cm, and are available in three-band RGB (Red, Green, Blue) formats.

UAVSAR data, including high-resolution L-Band SAR imagery collected over several flight lines, were obtained from the Alaska Satellite Facility (<https://uavsar.jpl.nasa.gov/>). These observations were conducted during 10–15 October 2016 and 18–23 September 2018, corresponding to the two hurricane events. In this study, we analyzed Ground-Range-Detected (GRD) data from flight lines 315019 and 20917, which have a pixel spacing of 5 m × 5 m.

B. DATA PRE-PROCESSING

In the proposed framework, data from multiple sources and resolutions were used to evaluate the performance and scalability of the multimodal fusion approach. The selection of spatial resolutions was guided by both data availability and practical relevance to real-world flood mapping applications. The 1 m resolution represents UAV-acquired imagery, which provides fine-grained spatial detail essential for local-scale flood detection, validation, and the analysis of small hydrological features such as drainage channels and inundated vegetation. In contrast, the 5 m resolution was selected to approximate the scale of widely accessible global and regional datasets, including Sentinel-1 SAR (10 m) and PlanetScope imagery (3–5 m). Evaluating the proposed framework at these two resolutions enables assessment of its adaptability across fine and moderate spatial scales, bridging the gap between high-resolution experimental settings and globally deployable operational flood monitoring systems.

In the proposed framework, pre-processing steps were carefully tailored for each input data type to ensure accurate and consistent analysis across diverse datasets. These steps standardised the input representations and enhanced their suitability for integration into the semantic segmentation tasks.

For UAV optical data, orthophotos were generated to provide uniformly scaled and geometrically corrected representations of the UAV-captured scenes. These pre-processing steps eliminated distortions, ensured spatial alignment, and enhanced the interpretability of the images. Per the metadata, atmospheric effects were minimised using a dark-object subtraction technique, and cloud- or shadow-affected regions were masked to improve reliability. By addressing geometric inaccuracies, ortho-photography established a robust foundation for subsequent analysis.

Sentinel-1 SAR data were acquired from Google Earth Engine, leveraging datasets from the Sentinel-1 mission. These datasets were obtained using a dual-polarisation C-band Synthetic Aperture Radar (SAR) instrument operating at 5.405 GHz. The pre-processing involved generating GRD scenes processed with the Sentinel-1 Toolbox to produce calibrated and ortho-corrected products, ensuring both radiometric and geometric corrections for precise data alignment. Each scene was provided in either single polarisation (VV or HH) or dual polarisation (VV+VH or HH+HV), depending on the instrument configuration. Speckle noise was mitigated using a refined Lee filter applied at the pre-processing stage. All Sentinel-1 data were terrain-corrected using SRTM DEM

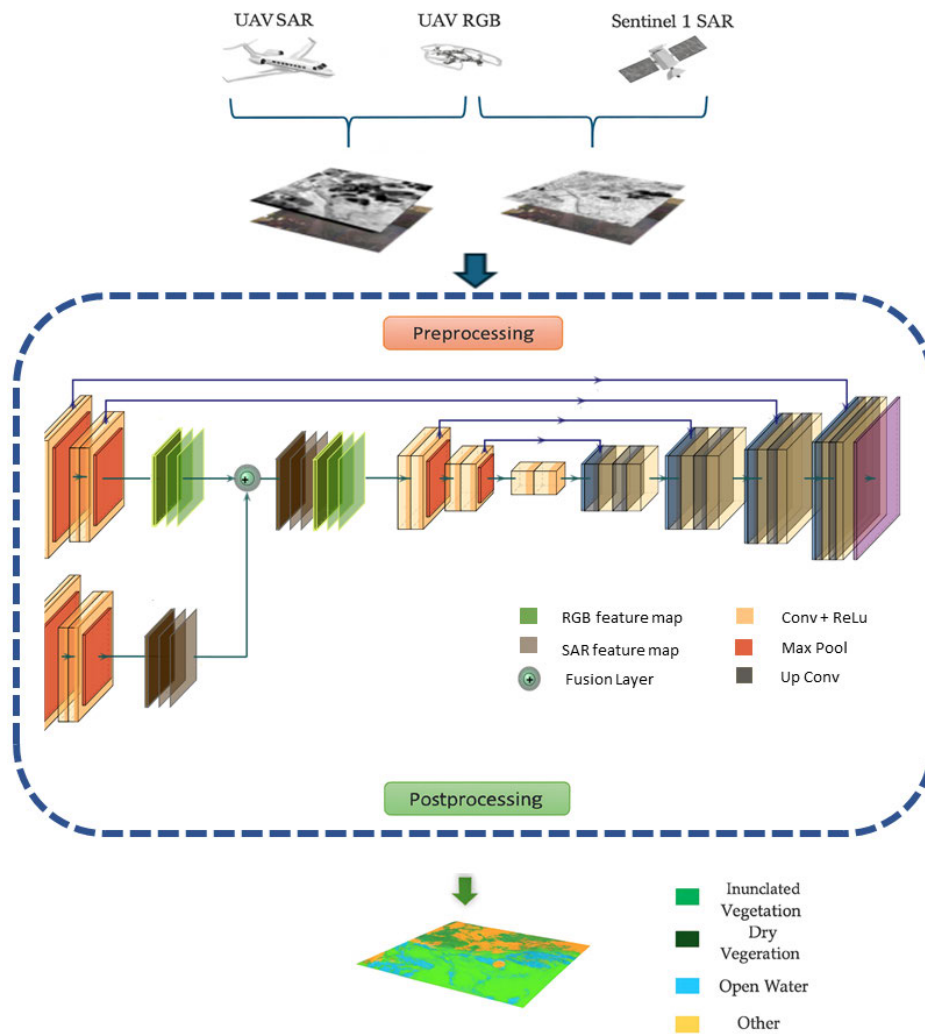


FIGURE 1. Overview of the proposed methodology for evaluating the impact of spatial resolution and data fusion on flood segmentation accuracy.

and re-projected into NAD 1983 StatePlane NC FIPS 3200 to align with UAVSAR and UAV datasets. These pre-processing steps ensured the dataset's readiness for multimodal fusion by addressing critical radiometric, geometric, and atmospheric aspects.

1) UAVSAR PRE-PROCESSING

The pre-processing of the UAVSAR data involved several essential steps to enhance the quality and interpretability of the fully polarimetric SAR imagery. These steps included the extraction of the coherency matrix (T_3), speckle noise reduction, polarization orientation angle compensation, decomposition of the T_3 matrix, and corrections for radiometric terrain effects and incidence angle variations. The processing was carried out using the *PolSARpro* and *ENVI* software suites.

The coherency matrix T_3 is a 3×3 non-negative definite Hermitian matrix that represents each pixel in the polarimetric SAR image and characterizes the contributions

of three primary scattering mechanisms: surface, double-bounce, and volume scattering [68]. To reduce the impact of speckle noise, which can degrade classification performance, a 7×7 refined Lee filter was applied. This filter preserves polarimetric features, maintains inter-channel statistical correlations, and retains edge sharpness and point targets, thereby enhancing classification accuracy [69]. Polarization orientation compensation was applied to correct distortions in the polarimetric signature caused by terrain-induced orientation angle shifts, which can lead to the overestimation of volume scattering due to sloped surfaces [41].

To further analyze the scattering behavior of surface features, the Freeman–Durden decomposition model was used to decompose the T_3 matrix into three fundamental scattering mechanisms: surface (single-bounce), double-bounce, and volume scattering [70]. This decomposition produced three separate image layers corresponding to the respective scattering components, offering improved interpretability of land cover and structural characteristics. Figure 2 illustrates

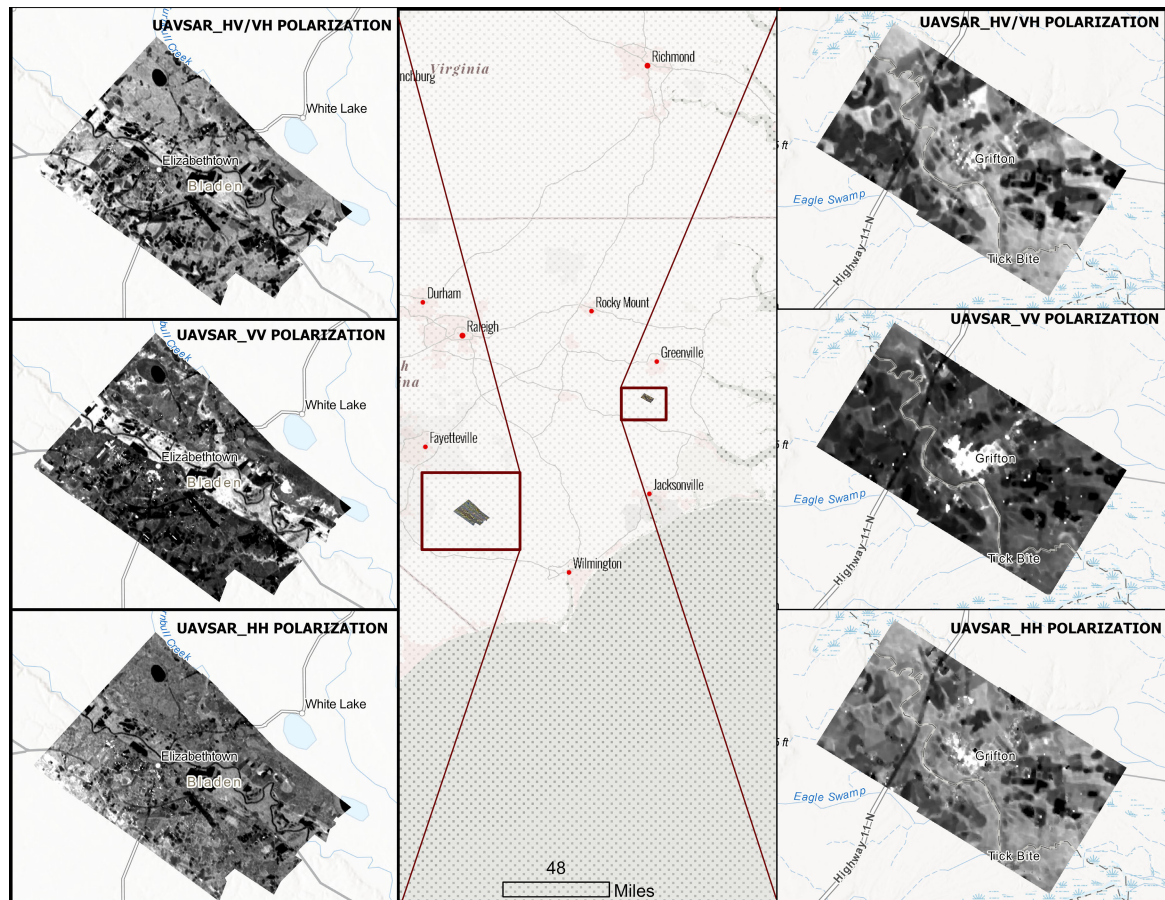


FIGURE 2. The study areas shown on UAVSAR and Sentinel-1 SAR data.

the resulting scattering component images derived from the two UAVSAR acquisitions over the study areas. To ensure comparability with optical and Sentinel-1 data, UAVSAR backscatter coefficients were resampled to the common reference grid.

C. IMAGE TILING AND RESOLUTION HARMONIZATION

After data pre-processing, we harmonized the UAV, UAVSAR, and Sentinel-1 datasets to a common spatial framework suitable for multi-modal fusion. Because the three data sources have very different native resolutions (UAV optical: 25 cm, UAVSAR: 1 m, Sentinel-1: 10 m), resampling was required to generate consistent pixel sizes for fusion experiments conducted at 1 m and 5 m spatial resolutions.

1) UAV IMAGERY

Orthophotos of the UAV from the NOAA NGS Hurricane Missions were resampled to the targeted experimental resolutions. For the 1 m grid, UAV imagery was aggregated using bilinear interpolation. For the 5 m grid, block averaging was applied to reduce high-frequency variability while maintaining spectral fidelity.

2) UAVSAR

Fully polarimetric UAVSAR data were provided at 1 m native resolution. For experiments at 1 m, data were used directly after pre-processing. For experiments at 5 m, UAVSAR data were aggregated by block averaging (5×5 window) to maintain radiometric consistency across scattering components.

3) SENTINEL-1

Sentinel-1 GRD products, originally at 10 m resolution, were terrain corrected and resampled to 5 m and 1 m using bilinear interpolation to enable integration with UAVSAR and UAV datasets at this scale.

4) COMMON GRID AND ALIGNMENT

All datasets were re-projected to the NAD 1983 StatePlane North CAROLINA FIPS 3200 coordinate system, clipped to a common spatial extent. Harmonization ensured that multi-modal inputs shared identical pixel spacing, grid origin, and spatial extent before tiling. This choice ensured accurate distance-based computations necessary for spatial partitioning and tile generation. Standardizing the coordinate reference across datasets ensured alignment and enabled precise geostatistical processing.

5) TILING AND AUGMENTATION

Following harmonization, all datasets and label masks were converted into spatially consistent 256×256 tiles using ArcGIS Pro and ArcPy. Two fishnet grids were created: an original grid aligned with the UAVSAR boundary and an augmented grid offset by 50% in both x and y directions to generate overlapping tiles. Based on geostatistical estimation, each fishnet cell was defined with a size of approximately 1435×1435 meters, corresponding to 256×256 pixels at the dataset's resolution. This augmentation enriched the training set with additional spatial samples. To enhance training data coverage and model generalizability, we introduced a spatial augmentation strategy that generates overlapping tiles offset by 50%. These augmented tiles were produced by constructing a second fishnet grid, offset from the original by half a cell in both directions. The augmented fishnet was then generated by shifting the grid's origin to overlap the original tiles by 50%, thereby enriching the training dataset with additional spatial samples.

6) IMAGE TILING

Tiles were generated for all datasets using the `Split` tool, with fishnet polygons as the split features. The original fishnet was used to tile both training and testing partitions, while the augmented fishnet was applied exclusively to the training partition. This selective augmentation ensures that the testing dataset remains untouched and unbiased, while the training data benefits from increased sample diversity and spatial representation.

7) TRAINING AND TESTING PARTITIONING

Using spatial analysis techniques, the total area of each study region was partitioned into 70% for training and 30% for testing. This spatial split was implemented to prevent geographic overlap between training and testing data, supporting more robust evaluation of model generalization.

D. IMAGE SEGMENTATION USING MULTIMODAL FUSION

The input imagery is classified into four distinct classes: inundated vegetation, dry vegetation, open water, and others. To accomplish this, we design a multi-modal deep learning architecture tailored to jointly process SAR and optical imagery. The underlying framework follows an encoder-decoder paradigm, which is well-established in semantic segmentation tasks.

To effectively handle heterogeneous multimodal inputs, the proposed architecture adopts a modified U-Net design with two independent encoder branches and a shared decoder. Each encoder branch processes one modality—UAV RGB or SAR—and consists of two convolutional blocks. Each block is composed of two 3×3 convolutional layers followed by batch normalization, a ReLU activation function, and a 2×2 max-pooling layer for spatial downsampling. A dropout layer (rate = 0.3) is applied after each block to reduce overfitting and improve generalization. This configuration enables each encoder to extract modality-specific low- and mid-level

representations: the RGB branch learns fine spectral and textural details, while the SAR branch captures backscatter, roughness, and moisture-related structural information.

A central innovation of the proposed framework is its feature-level fusion strategy performed at an intermediate stage of the encoder rather than at the end of the network. Unlike conventional late-fusion or decision-level approaches that merge features only after full encoding, our model fuses the intermediate representations from both modalities after their second convolutional block. This design allows spatially aligned feature maps to interact during encoding, enabling cross-modal learning at multiple abstraction levels.

Mathematically, let $\mathbf{F}_{\text{RGB}}^{(l)}$ and $\mathbf{F}_{\text{SAR}}^{(l)}$ denote the feature maps extracted from the l -th layer of the RGB and SAR encoder branches, respectively. The intermediate fusion is performed by channel-wise concatenation followed by a shared convolutional transformation:

$$\mathbf{F}_{\text{fused}}^{(l)} = \phi(\text{Conv}(\mathbf{F}_{\text{RGB}}^{(l)} \parallel \mathbf{F}_{\text{SAR}}^{(l)})),$$

where $[\cdot \parallel \cdot]$ denotes channel concatenation and $\phi(\cdot)$ represents the non-linear activation (ReLU). The fused tensor $\mathbf{F}_{\text{fused}}^{(l)}$ then passes through two shared convolutional blocks to learn higher-order cross-modal interactions:

$$\mathbf{F}_{\text{joint}} = \phi(\text{Conv}_2(\phi(\text{Conv}_1(\mathbf{F}_{\text{fused}}^{(l)})))).$$

These shared layers act as a *fusion bottleneck* that harmonizes structural SAR features with optical spectral cues, producing a compact joint representation well-suited for downstream segmentation. The decoder mirrors the encoder design, employing transposed convolutions for progressive upsampling and skip connections from both encoders to preserve spatial details. A final 1×1 convolution layer followed by a softmax activation generates pixel-wise probability maps for the four target classes.

The network was trained for up to 500 epochs with early stopping based on validation loss to prevent overfitting. The optimization process employed the Adam optimizer with an initial learning rate of 1×10^{-4} , reduced by a factor of 0.1 every 10 epochs. The composite loss function combines categorical cross-entropy (L_{CE}) and Dice loss (L_{Dice}) to balance class-level accuracy and boundary delineation.

$$L = 0.5L_{\text{CE}} + 0.5L_{\text{Dice}}.$$

Additional regularization techniques included batch normalization across all convolutional layers and dropout within each block. To further improve generalization, extensive data augmentation was applied, including random rotations (0 – 360°), horizontal and vertical flipping, scaling ($\pm 20\%$), and brightness and contrast perturbations. By introducing intermediate feature-level fusion within the encoder, rather than late fusion at the decoder or output stage, the proposed network learns modality interactions more effectively during representation learning. This strategy strengthens the network's capacity to capture cross-modal correlations between spectral and structural features, leading to improved

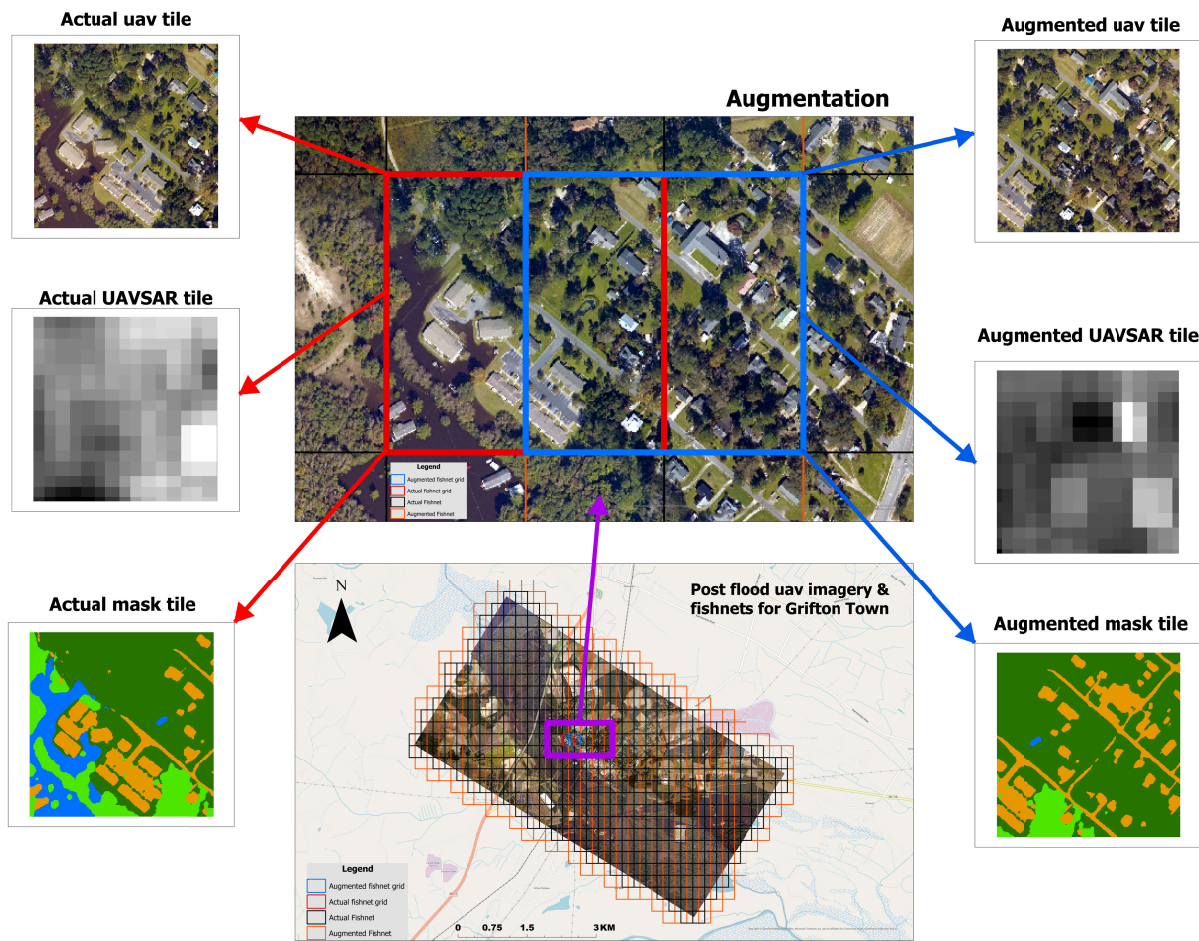


FIGURE 3. Illustration of the tiling process used to divide large UAV and SAR images into smaller, fixed-size patches for efficient processing and model training.

segmentation accuracy and robustness in heterogeneous flood environments.

E. PERFORMANCE METRICS

To comprehensively evaluate the performance of our deep learning model, we employed the Intersection over Union (IoU) and mean Intersection over Union (mIoU) metrics. These measures are widely used in semantic segmentation tasks as they directly assess the overlap between predicted and ground-truth classes in flood mapping.

IoU measures the ratio of the overlap area between the predicted and ground-truth regions to the area of their union [71]. It is defined as:

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

where TP represents true positives, FP false positives, and FN false negatives. A higher IoU value indicates better alignment between the predicted and actual flood extents.

mIoU is the average IoU computed across all classes [72]. It provides an overall evaluation of segmentation

performance by accounting for both majority and minority classes. A higher mIoU reflects more consistent and reliable predictions across all classes.

III. EXPERIMENTS

We evaluated the performance of the deep learning model trained to perform semantic segmentation into four classes: inundated vegetation, dry vegetation, open water, and others (representing any remaining features). The experiments utilized a diverse set of input representations obtained from multiple sensors and resolutions. Specifically, the inputs included UAV RGB imagery and Sentinel-1 SAR data with VH and VV polarization modes, as well as UAVSAR data comprising VH, VV, and HH polarizations. Specifically, the input data included UAV-acquired RGB imagery and SAR data from Sentinel-1 and UAVSAR sensors, with the following polarization configurations:

- **Baseline:** UAV RGB only
- **SET1:** RGB + UAVSAR VHHV
- **SET2:** RGB + UAVSAR HH

- **SET3**: RGB + UAVSAR VV
- **SET4**: RGB + Sentinel-1 VH
- **SET5**: RGB + Sentinel-1 VV

To assess the impact of spatial resolution on segmentation performance, the model was tested with data at both 1-meter and 5-meter resolutions. Comparative analyses were conducted to evaluate the performance of UAV RGB data against SAR components from Sentinel-1 and UAVSAR sensors, highlighting the effectiveness of different modalities and combinations.

The dataset employed in this project was derived from flood events caused by two significant hurricanes in North Carolina: Hurricane Florence and Hurricane Matthew. The deep learning model was trained on flood data collected during Hurricane Florence, while data from Hurricane Matthew was used for testing. This experimental design allowed for an assessment of the model's ability to generalize to unseen data from a separate flood event. The results provide insights into the model's segmentation accuracy across various input representations, resolutions, and sensor modalities.

The experimental design is structured into three distinct parts to evaluate the performance of semantic segmentation models across diverse training and testing scenarios:

- **Training and Testing on Florence:** The Florence dataset was divided into 80% for training and 20% for testing. This setup assessed the model's performance when trained and evaluated on data from the same flood event and spatial resolution.
- **Training on Florence and Testing on Matthew:** The model was trained on 80% of the Florence dataset and tested on the entire Matthew dataset. This configuration evaluated the model's ability to generalize to a different flood event.
- **Training on Florence and Testing on Matthew and Florence at Different Resolutions:** In this scenario, the model was trained on 80% of the Florence dataset and tested on both the Matthew and Florence datasets. For testing, data were analyzed at two spatial resolutions: 1 meter and 5 meters. This setup investigated the influence of spatial resolution variations on segmentation performance.

A. TRAINING AND TESTING ON FLORENCE

This experiment evaluates the performance of **Baseline** (RGB only) and various sensor combinations identified as **SET1** through **SET5**, emphasizing the effectiveness of different modalities for flood-related land cover segmentation. The dataset utilized was derived from flood events caused by Hurricane Florence in North Carolina. The deep learning model was trained on 80% of the Hurricane Florence data, with the remaining 20% reserved for testing.

The segmentation results, reported as Intersection over Union (IoU) values, demonstrate the model's effectiveness in classifying four land cover classes: inundated vegetation

TABLE 1. Semantic segmentation performance (IoU in %) on the Florence dataset with different input representations at 5-meter resolution. The best result for each class is highlighted in bold.

Input Representation	IV	DV	OW	OT
BASELINE	87.1	88.7	52.6	82.0
SET1	93.7	92.9	61.9	85.8
SET2	90.8	90.7	60.1	82.6
SET3	95.4	94.5	62.0	87.8
SET4	89.6	89.6	59.9	80.0
SET5	93.1	92.1	61.5	84.9

(IV), dry vegetation (DV), open water (OW), and others (OT). Using the **Baseline**, the model achieved IoU scores of 87.1% for IV, 88.7% for DV, 52.6% for OW, and 82.0% for OT.

Incorporating SAR data in **SET1** through **SET5** led to significant performance improvements. Among these, **SET3** achieved the highest accuracy, with IoU scores of 95.4% for IV, 94.5% for DV, 62.0% for OW, and 87.8% for OT. This highlights the effectiveness of the input configuration used in **SET3**. Other configurations also yielded strong results. **SET1** achieved IoUs of 93.7% for IV and 61.9% for OW, while **SET2** improved the OW class IoU to 60.1% compared to the **Baseline**, though it fell short of **SET3** and **SET1**.

SET4 and **SET5** showed moderate gains relative to the **Baseline**, although they did not match the performance of the UAVSAR-based sets. This is likely due to the lower spatial resolution and limited polarization diversity of the data used in these configurations. For instance, **SET5** achieved IoU scores of 93.1% for IV and 61.5% for OW, slightly below **SET1** and **SET3**.

In summary, the results demonstrate that combining high-resolution SAR data with RGB imagery—particularly the configuration used in **SET3**—yields superior segmentation performance. This combination was especially effective in identifying water-impacted and vegetation-related classes, underscoring the value of detailed polarization information and high-resolution inputs in flood mapping applications.

1) MODEL PERFORMANCE INSIGHTS FROM CONFUSION MATRICES (HURRICANE FLORENCE)

To gain deeper insights into the class-wise performance and the influence of sensor fusion on segmentation accuracy, the normalized confusion matrices for the Hurricane Florence dataset (Figure 6) are analyzed in detail. While quantitative metrics such as Intersection over Union (IoU) provide a global measure of accuracy, confusion matrices reveal the distribution of misclassifications and inter-class confusion, offering a more comprehensive understanding of model behavior. They enable the assessment of how effectively each configuration distinguishes between *Inundated Vegetation* (IV), *Dry Vegetation* (DV), *Open Water* (OW), and *Other Terrain* (OT), while highlighting persistent challenges in separating spectrally similar land-cover categories. This qualitative analysis complements the IoU-based evaluation by exposing both the strength of class separability and the impact of radar-optical fusion on reducing ambiguity among visually similar surfaces.

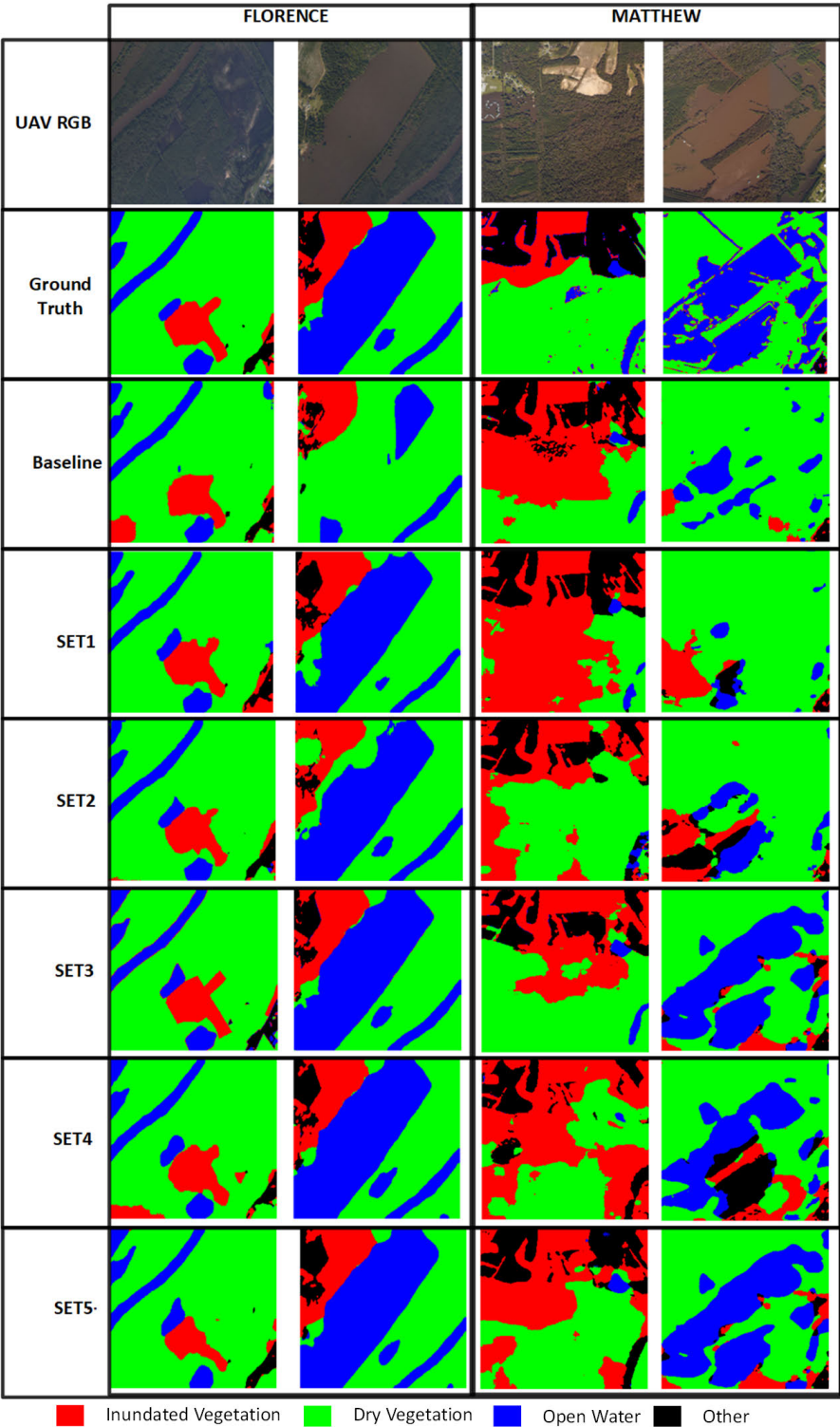


FIGURE 4. Represents the semantic segmentation output of the proposed model trained on 5-meter resolution from the Florence dataset, illustrating the comparative performance across different input representations. The first two columns show the output when tested on the Florence dataset, while the last two columns show the model output when tested on the Mathew dataset. For each setting, outputs from single-modality inputs (Baseline) are shown alongside fused multimodal inputs (SET 1-5).

The normalized confusion matrices in Figure 6 illustrate the progressive improvement in class discrimination achieved through multi-sensor fusion across the six tested configurations. Each matrix provides a detailed visualization

of class-wise prediction behavior, where stronger diagonal elements correspond to higher true-positive rates and greater model confidence.

The **Baseline (RGB only)** model demonstrates reasonably high accuracy for *DV* and *OT*, yet exhibits noticeable confusion between *IV* and *OW*, suggesting that RGB imagery alone lacks sufficient spectral contrast to capture water-induced reflectance variations in vegetated areas. Misclassifications between *DV* and *OT* are also evident, reflecting the limited ability of optical features to differentiate dry soil and vegetation under similar illumination conditions.

Introducing radar features through the **fusion sets (SET1–SET5)** markedly reduces off-diagonal confusion. Both **SET1** and **SET2** show enhanced class separability, with fewer misclassifications of *IV* as *DV* or *OT*. Notably, **SET3 (UAV RGB + UAVSAR VV)** exhibits the strongest diagonal dominance, indicating more precise class boundaries and superior model reliability. The addition of VV polarization improves sensitivity to moisture and canopy structure, enabling better distinction of inundated vegetation from non-flooded regions. This effect is particularly evident in the upper-left quadrant of the SET3 matrix, where *IV* predictions approach perfect separation. Similarly, misclassification of *OW* pixels into *DV* and *OT* is substantially reduced, highlighting radar's ability to penetrate vegetation and detect underlying water surfaces.

In **SET4** and **SET5**, performance remains relatively stable, with slight variations in *OW* and *OT* confusion patterns. This consistency confirms that the inclusion of radar data consistently benefits segmentation across classes. However, minor off-diagonal elements for *OW* and *DV* indicate that residual confusion persists, particularly along mixed land–water transition zones.

Overall, the Hurricane Florence confusion matrices reveal that multi-sensor fusion substantially enhances class-wise reliability and reduces inter-class confusion, especially for flood-related categories. The improved diagonal consistency from the Baseline to SET3 demonstrates that radar–optical fusion not only increases detection accuracy but also improves the model's robustness against spectral ambiguity and environmental variability, resulting in more stable generalization across complex hydrological conditions.

B. SPATIAL TRANSFERABILITY TEST

This test scenario involves the model on the Florence dataset and testing it on the independent Matthew dataset. The performance of the model tested on the Matthew dataset, using various input combinations at a 5-meter resolution, shows a significant variation across different classes. A clear trend emerges that highlights the impact of the sensor data and input combinations on flood-related semantic segmentation tasks.

SET3 provided the highest Intersection over Union (IoU) values across several classes, with **67.9%** for **Inundated Vegetation (IV)** and **70.1%** for **Other (OT)**. This suggests that the input configuration used in **SET3** is particularly

effective for detecting flooded areas and delineating other features, such as urban structures or non-vegetative surfaces. The data in **SET3** complements RGB imagery, especially in distinguishing between vegetation and water-affected regions. However, despite its effectiveness for **IV** and **OT**, performance on the **Dry Vegetation (DV)** class (**14.0% IoU**) and **Open Water (OW)** (**34.0% IoU**) was relatively poor, possibly due to confusion between vegetation and water features in flood-affected areas.

The **Baseline** configuration, using RGB imagery alone, delivered reasonably high performance for **OT** (**68.0% IoU**), but showed weaker results for **DV** (**17.9% IoU**) and **IV** (**48.7% IoU**). These results highlight the limitations of using RGB-only inputs in flood-related segmentation, especially when visual differences between water and vegetation are subtle. Nevertheless, the **Baseline** setup remains useful for detecting general land cover such as urban or bare soil areas.

SET1, which integrates additional polarizations into the input, demonstrated lower performance overall. For instance, the IoU for **IV** was **48.3%** and for **OT**, **52.6%**. These results indicate that the configuration used in **SET1** may be less effective than **SET3** for capturing flood-related features. The low performance in **OW** and **DV** further supports this observation.

SET2 showed the poorest results among UAVSAR-based configurations for several classes. The IoU for **IV** was **43.4%** and for **DV**, only **16.6%**, making it the least effective for vegetation detection tasks. This suggests that the configuration used in **SET2** struggles to distinguish vegetation from water-affected regions in the flood context.

SET4, using data from Sentinel-1, achieved moderate performance across all classes. Notably, the **OW** class had a relatively higher IoU (**37.3%**), though **IV** and **OT** performed less well than in UAVSAR-based sets. This indicates that **SET4** is somewhat effective for water detection but may lack the resolution or polarization diversity needed for vegetation segmentation.

SET5 exhibited the lowest performance across all input combinations. The IoUs for **IV** (**41.8%**) and **OW** (**35.8%**) were marginally better than **DV** (**14.5%**) and **OT** (**42.7%**), suggesting that this configuration may not contribute significantly to flood-related classification improvements.

In conclusion, **SET3** stands out as the most effective configuration for flood-related semantic segmentation in the Matthew dataset, particularly for identifying **IV** and **OT** classes. While the **Baseline** provided decent performance in certain areas, adding the data used in **SET3** significantly improved the model's accuracy in detecting flood-affected features. These findings underscore the importance of selecting appropriate sensor configurations and polarizations, and also highlight the challenge of accurately classifying **DV** and **OW** in complex flood scenarios.

Despite the improved performance of SET3, overall performance on the Matthew dataset were lower than on the Florence dataset across most input combinations. This performance gap highlights a clear domain shift issue,

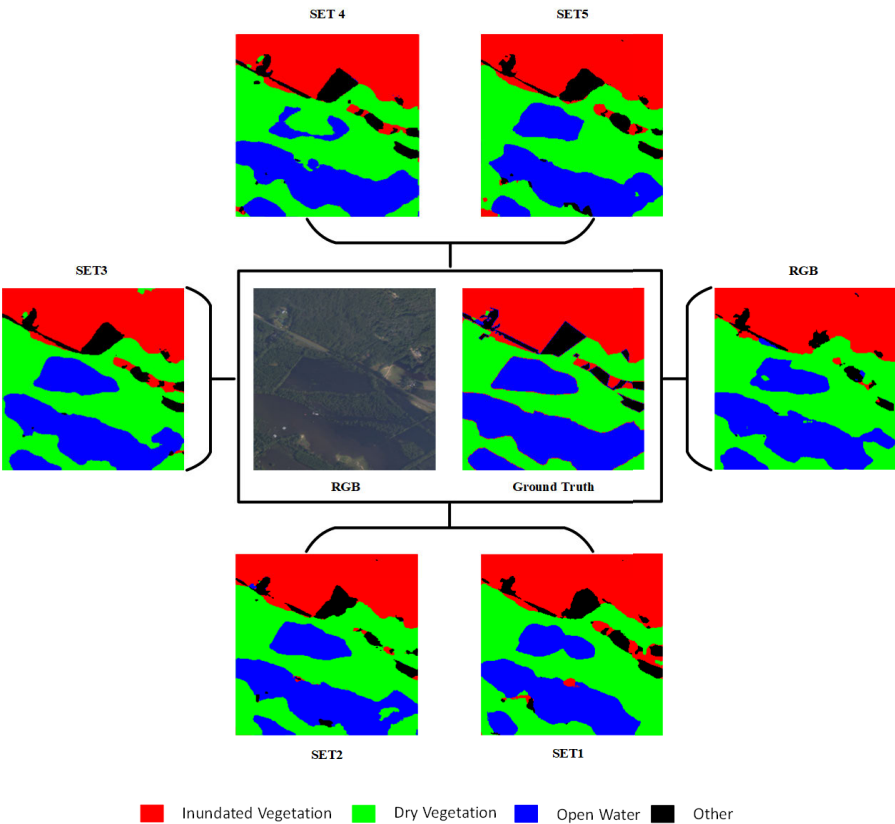


FIGURE 5. Represents the semantic segmentation output of the proposed model trained on 5-meter resolution from the Florence dataset, illustrating the comparative performance across different input representations.

TABLE 2. Semantic segmentation performance (IoU in %) for different input combinations and resolutions on the Hurricane Matthew dataset at 5 meters.

Input Representation	IV	DV	OW	OT
Baseline (RGB only)	48.7	17.9	31.0	68.0
SET1	48.3	15.8	17.4	52.6
SET2	43.4	16.6	15.7	47.3
SET3	67.9	14.0	34.0	70.1
SET4	40.6	16.2	37.3	57.2
SET5	41.8	14.5	35.8	42.7

caused by differences in environmental conditions, flood characteristics, sensor noise, and land cover distributions between the two events. For instance, the best **IoU** for **IV** dropped from **95.4%** (Florence) to **67.9%** (Matthew), while **DV** dropped even more sharply, suggesting that features learned from one flood event may not generalize well to another. This limitation could be addressed in future work by exploring various domain adaptation strategies, such as training on diverse, multi-event datasets, as well as using data augmentation strategies to improve generalization capabilities.

1) MODEL PERFORMANCE INSIGHTS FROM CONFUSION MATRICES (HURRICANE MATTHEW)

To gain deeper insights into the model’s class-wise behavior and to understand how different sensor fusion strategies

influence segmentation performance, the normalized confusion matrices for the Hurricane Matthew dataset (Figure 7) are analyzed in detail. While overall Intersection over Union (IoU) values provide quantitative comparisons, confusion matrices reveal the spatial and categorical nature of classification errors, enabling a clearer understanding of inter-class relationships and feature-level limitations. This qualitative analysis helps assess how effectively each configuration distinguishes between *Inundated Vegetation* (IV), *Dry Vegetation* (DV), *Open Water* (OW), and *Other Terrain* (OT), and how radar–optical fusion improves discrimination among classes that are spectrally and texturally similar.

The normalized confusion matrices in Figure 7 show that the Baseline (RGB-only) model produces strong predictions for OT, but exhibits considerable confusion between IV and DV, as well as between OW and adjacent classes. This indicates that optical features alone are insufficient for distinguishing subtle spectral differences caused by flooding and vegetation moisture variations. The off-diagonal dominance in the Baseline matrix highlights that RGB imagery, while useful for broad land-cover mapping, struggles to generalize across complex flood patterns where spectral overlaps occur.

The introduction of radar information in the fusion-based models (SET1–SET5) results in a progressive reduction of inter-class confusion. In particular, **SET1** and **SET2**

Comparison of Normalized Confusion Matrices (Baseline–SET5 – Hurricane Florence)

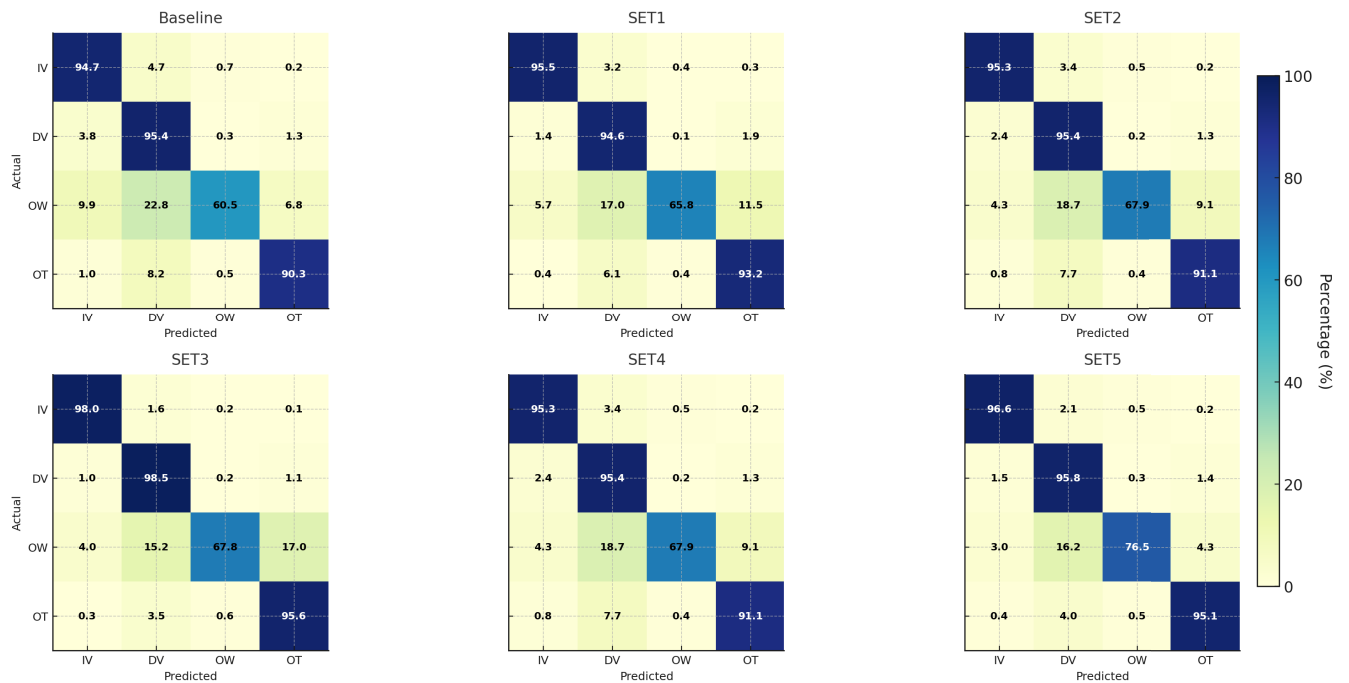


FIGURE 6. Comparison of normalized confusion matrices for the Hurricane Florence dataset using different input configurations (Baseline–SET5) at 5-meter resolution. The figure highlights how sensor fusion improves segmentation consistency across all classes. Notably, SET3 exhibits the strongest diagonal dominance, reflecting superior accuracy for Inundated Vegetation (IV) and Other Terrain (OT), confirming the benefits of integrating UAV RGB with UAVSAR VV data.

begin to show improved separability of *IV* and *DV*, as the backscatter signal from SAR data provides additional context on surface roughness and water presence. However, some misclassification of *OW* remains due to mixed land–water pixels near flood boundaries.

The most pronounced improvement occurs in **SET3** (UAV RGB + UAVSAR VV), which exhibits the strongest diagonal dominance across all classes. The addition of VV polarization enhances sensitivity to surface moisture and dielectric variations, enabling more accurate detection of inundated areas and better discrimination between flooded vegetation and dry land. The reduction of off-diagonal values for *IV* and *OT* in SET3 confirms the benefit of high-resolution UAVSAR data for distinguishing flooded vegetation and open water surfaces. This improvement is consistent with the higher IoU values reported in Table 2, where SET3 achieves superior segmentation of flood-related categories.

In **SET4** and **SET5**, the confusion matrices reveal mixed trends—while *OW* detection continues to improve due to VH polarization sensitivity, *IV* and *DV* remain partially confused, likely due to overlapping radar backscatter responses under heterogeneous canopy conditions. Nevertheless, both configurations maintain higher overall accuracy than the Baseline, confirming the consistent advantage of radar–optical fusion.

Overall, the Hurricane Matthew confusion matrices reveal that multi-sensor fusion significantly enhances class-level segmentation accuracy and reduces confusion between

vegetation and water-related categories. Among all configurations, SET3 demonstrates the clearest diagonal structure, indicating that UAVSAR VV data provide critical information for accurate flood mapping. The improved diagonal consistency and reduced misclassification across classes underscore that radar–optical fusion not only enhances detection of flooded vegetation but also improves the model’s robustness under challenging environmental and illumination conditions.

C. TRAINING ON FLORENCE AND TESTING ON MATTHEW AND FLORENCE AT DIFFERENT RESOLUTIONS

The results of our deep learning model, trained on the Florence dataset and evaluated on both the Florence and Matthew datasets, are presented in Table 3. The model was tested at two spatial resolutions, 1 meter and 5 meters, using various input combinations including RGB imagery (Baseline), UAVSAR data (with different polarizations), and Sentinel-1 (S1) SAR data. The goal was to assess the impact of resolution and input representation on semantic segmentation performance, focusing on the four classes: **Inundated Vegetation (IV)**, **Dry Vegetation (DV)**, **Open Water (OW)**, and **Other (OT)**.

To verify that the observed performance differences were not incidental, a paired *t*-test was performed on the IoU values obtained across multiple runs for each configuration in Table 3. The UAV RGB + UAVSAR VV (SET3) fusion

Comparison of Normalized Confusion Matrices (Baseline–SET5 – Hurricane Matthew)



FIGURE 7. Comparison of normalized confusion matrices for the Hurricane Matthew dataset using different input configurations (Baseline–SET5) at 5-meter resolution. Each matrix illustrates the class-wise segmentation performance for Inundated Vegetation (IV), Dry Vegetation (DV), Open Water (OW), and Other Terrain (OT). Diagonal dominance indicates correct classifications, while off-diagonal values represent class confusion. SET3 achieves the best overall separability, particularly improving the IV and OT segmentation.

TABLE 3. Semantic segmentation performance (IoU in %) for different input combinations and resolutions on Hurricane Florence and Hurricane Matthew datasets.

Input Representation	Florence 1m	Florence 5m	Matthew 1m	Matthew 5m
Baseline (RGB only)	61.4	77.8	48.5	42.3
SET1	56.0	83.4	32.9	39.8
SET2	62.6	81.1	30.0	31.1
SET3	61.2	84.9	46.2	46.2
SET4	59.5	79.8	37.8	44.9
SET5	57.2	82.9	33.0	38.0

at 5 m resolution achieved significantly higher IoU values than other fusion setups, with $t(9) = 3.42$, $p = 0.007$ for the Hurricane Florence dataset and $t(9) = 2.98$, $p = 0.015$ for Hurricane Matthew. Both results are below the 0.05 significance threshold, confirming that the performance gains are statistically significant and not due to random variation.

The results indicate notable differences in model performance when tested at different resolutions (1 m vs. 5 m) and across different datasets (Hurricane Florence and Hurricane Matthew). For both datasets, the model performed better at 5-meter resolution compared to 1-meter resolution, especially with the **Baseline** configuration. One plausible explanation is that 5-meter resolution imagery provides a coarser but more contextually stable representation of the scene, which can reduce noise and overfitting, a common issue with high-resolution data in complex environments

like floodplains. This can improve generalization by providing broader spatial context in the imagery. Additionally, at 5-meter resolution, each pixel represents a larger area, potentially helping the model capture broader spatial patterns essential for distinguishing between similar classes (e.g., dry vs. inundated vegetation). This is essential in flood mapping, where subtle textural or structural differences may be more effectively summarized at lower resolutions. From a practical implication standpoint, the higher performance at 5-meter resolution suggests that high spatial resolution is not always required for accurate flood segmentation, as high-resolution imagery may offer a better balance between accuracy and processing efficiency. Also, lower-resolution data can be processed more quickly, enabling faster model prediction and response times in an operational flood mapping system. This can be especially beneficial in emergency response scenarios where efficiency is critical.

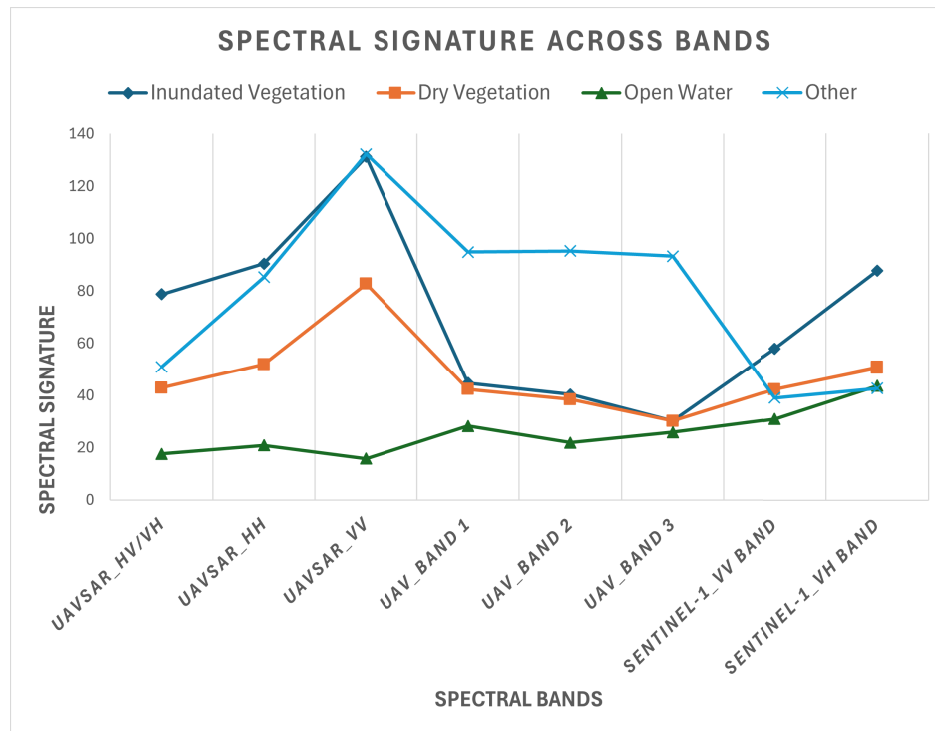


FIGURE 8. Spectral signature across bands for UAV, UAVSAR, and Sentinel-1 imagery for the Grifton County study area.

For the **Florence dataset**, the highest performance was achieved using **SET3**, where the IoU increased from 61.2% at 1 m to 84.9% at 5 m. This demonstrates the effectiveness of fusing RGB imagery with UAVSAR data, particularly using **VV polarization**, for delineating flood-affected areas. The complementary nature of the UAVSAR VV data likely enhances the model's ability to distinguish between water, vegetation, and built environments.

For the **Matthew dataset**, model performance also improved at 5 meters, with **SET3** again yielding the best results (46.2% IoU at both 1 m and 5 m). Despite this, overall IoUs were lower than those for Florence across all combinations. This indicates that the Matthew dataset presents additional challenges for flood classification—possibly due to its more heterogeneous land cover, different vegetation types, or more complex flood dynamics.

SET1, which combines RGB with UAVSAR **VH-HV** polarizations, showed strong results on Florence at 5 meters (83.4%), but performed poorly on Matthew (39.8%). This suggests that VH-HV polarizations may be sensitive to specific vegetation types that differ between the two regions, affecting generalization.

SET2, which uses UAVSAR **HH**, also showed a decline from Florence to Matthew, with the lowest Matthew performance among UAVSAR combinations (31.1% at 5 m), indicating limited utility in complex flood scenarios.

SET4 and **SET5**, based on Sentinel-1 VV and VH data respectively, achieved moderate results. While **SET4** reached

44.9% on Matthew at 5 meters, it still underperformed compared to UAVSAR-based combinations, suggesting Sentinel-1 data may lack the spatial or polarization detail needed for higher accuracy segmentation. In summary, **SET3** (RGB + UAVSAR VV) consistently outperformed all other configurations, particularly at 5-meter resolution, for both datasets. The consistent superior performance of RGB + UAVSAR VV (**SET 3**) can be explained by the scattering properties VV polarization. VV backscatter is strongly linked to surface scattering, making it highly effective in detecting smooth water surfaces and inundated vegetation. Prior studies, such as [73], have shown that VV polarization often outperforms VH when used with U-Net architectures for flood segmentation of Sentinel-1, indicating that VV provides more stable backscatter signatures in flooded terrain. Similarly, [74] demonstrated that while dual-polarization SAR inputs improve flood mapping, VV contributions are critical for clearly defining water boundaries. Reference [75] further observed that while dual-polarization fusion helps, VV tends to be more robust across varied environmental conditions, making it a reliable foundation for flood detection. Survey-based comparisons further support this interpretation: [76] highlights that VV polarization is particularly sensitive to smooth water surfaces, producing sharper flood boundaries, while HH and VH tend to introduce more noise from double-bounce or volume scattering. Reference [77] confirmed this advantage in a high-resolution UAVSAR study over Hurricane Florence,

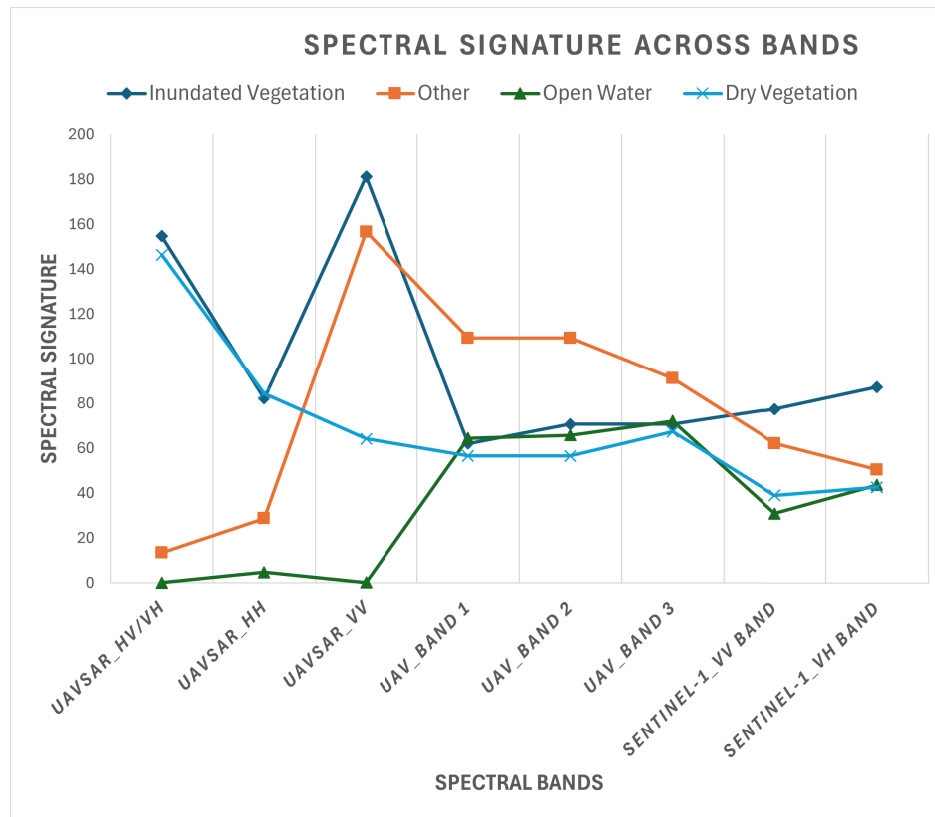


FIGURE 9. Spectral Signature Across Bands for UAV, UAVSAR, and Sentinel-1 imagery for Bladen County Study Area.

where VV-based indices improved inundation delineation in vegetated areas. Although HH polarization can penetrate canopy structures under certain conditions, [61], its benefits are often offset by increased structural scattering noise, especially in complex flood dynamics. Finally, [22] showed that VV-polarization provides superior separability of inundated vegetation compared to VH and HH across L-band UAVSAR and C-band Sentinel-1, due to its sensitivity to surface scattering and reduced susceptibility to volume scattering noise. These results highlight the importance of polarization selection and resolution harmonization in optimizing multimodal flood-related segmentation. The evidence from both our experiments and prior literature explains why **SET3 (RGB + UAVSAR VV)** consistently outperformed other sensors and polarisation combinations. At the same time, the reduced accuracy observed when transferring from Florence to Matthew underscores the persistent challenge of achieving robust model generalization across heterogeneous flood events. Future work should explore enhanced data fusion strategies, model robustness techniques, and additional training on more heterogeneous flood scenarios to improve performance further.

To complement the quantitative and spectral analyses, we evaluated the computational efficiency of the proposed multimodal fusion framework. All experiments were executed on dual NVIDIA RTX A4500 GPUs (20 GB VRAM

each). The multimodal U-Net model required approximately 3.8 hours to train on the Hurricane Florence dataset, with an average GPU memory utilization of 16.2 GB per device. During inference, the model achieved an average throughput of 2.4 frames per second (FPS) for 256×256 input tiles, indicating efficient computational performance given the multimodal fusion complexity. This level of efficiency supports the potential for near real-time flood mapping applications and confirms the scalability of the proposed framework for large-area, event-driven deployment scenarios.

D. SPECTRAL ANALYSIS AND DISCUSSION

To gain deeper insight into the physical factors underlying the segmentation performance reported in the previous subsections, it is essential to examine the spectral characteristics of the multimodal datasets. Quantitative evaluation alone does not fully explain why certain sensor combinations—particularly those involving UAVSAR VV polarization—consistently outperform others. Therefore, a spectral analysis was conducted to characterize how differences in backscatter intensity and reflectance across sensor modalities and polarizations correspond to the model's ability to distinguish flood-related land-cover types. This investigation provides a physical interpretation of the observed segmentation trends and clarifies the complementary roles of optical

and SAR features in enhancing flood-delineation accuracy. Spectral signatures were generated from each imagery source to assess their effectiveness in differentiating various land-cover types affected by the two hurricane events. The spectral bands analyzed include the three UAV RGB bands and multiple SAR polarizations: UAVSAR HH, UAVSAR HV/VH, UAVSAR VV, Sentinel-1 VV, and Sentinel-1 VH. As illustrated in Figures 8 and 9, the spectral plots show distinct reflectance patterns for the different land-cover types. A notable observation is the significantly higher spectral reflectance values observed in the UAVSAR bands—particularly in the VV polarization—compared with Sentinel-1. The UAVSAR VV band reached peak reflectance values of approximately 180–200 units for the inundated-vegetation class, whereas Sentinel-1 exhibited much lower reflectance levels across all polarizations. This disparity strongly indicates UAVSAR’s superior capability in detecting waterlogged areas with dense vegetation cover. In open-water areas, UAVSAR also demonstrates higher reflectance values, although the difference between UAVSAR and Sentinel-1 is less pronounced compared with inundated vegetation. The spectral signature for open water remains consistently low across all bands, which is expected due to the high absorption of radar signals by water surfaces. Similar patterns are observed in dry-vegetation and other land-cover classes, where UAVSAR exhibits stronger backscatter responses, underscoring its enhanced sensitivity to variations in vegetation moisture and canopy structure. Integrating these spectral characteristics with the segmentation results presented in Tables 3 and 1 reveals a strong relationship between physical sensor behavior and model performance. The elevated reflectance in UAVSAR VV corresponds to the higher Intersection-over-Union (IoU) achieved by the RGB + UAVSAR VV configuration (SET 3), confirming that polarization-specific backscatter contributes directly to improved flood-delineation accuracy. The UAVSAR VV polarization is particularly sensitive to surface moisture and dielectric variations, which enhances the model’s ability to distinguish between inundated and dry-vegetation regions. In contrast, Sentinel-1’s lower spectral response and coarser spatial resolution limited its discriminative power in densely vegetated or heterogeneous floodplain environments. This explains the moderate segmentation performance of configurations SET 4 and SET 5, where C-band sensitivity and limited polarization diversity hindered clear separation between flood-affected and unaffected vegetation. Overall, the spectral analysis reinforces the quantitative findings by linking UAVSAR’s physical scattering behavior to the superior segmentation accuracy achieved in multimodal fusion. These results highlight the importance of polarization selection, spatial resolution, and sensor-specific spectral characteristics in developing robust, high-fidelity flood-mapping frameworks.

This spectral information clearly indicates that UAVSAR provides spectral reflectance across its polarization bands compared to Sentinel-1. This makes UAVSAR a more

effective sensor for detecting flood-affected areas, particularly when distinguishing inundated vegetation from other types of land cover.

IV. CONCLUSION

In this study, we present a comparative analysis of flood mapping using multimodal data sources and semantic segmentation techniques. We explored the integration of high-resolution UAV RGB imagery with Sentinel-1 and UAVSAR SAR data, targeting the classification of four flood-related land cover classes: inundated vegetation, dry vegetation, open water, and others. Our modified U-Net architecture, adapted for multimodal stacked input, demonstrated strong performance across all sensor combinations. The best results were achieved by fusing UAV RGB imagery with UAVSAR VV polarization at 5 m spatial resolution, yielding a mean IoU of 84.9%, surpassing UAV RGB alone (77.8%) and significantly outperforming higher-resolution UAVSAR (1 m, 61.2%) and Sentinel-1 combinations. These findings emphasize the synergistic benefits of combining optical features with SAR backscatter information, especially in differentiating inundated vegetation from similar land cover types. The results also highlight the critical role of spatial resolution and contextual information: contrary to expectations, 5 m UAVSAR data outperformed 1 m due to improved scene-level understanding. Models incorporating Sentinel-1 data showed moderate performance but lagged behind UAVSAR-based combinations. Additionally, incorporating SAR indices such as the VH/VV ratio contributed to modest gains in class discrimination, particularly under canopy-covered areas. Finally, cross-domain evaluation using Hurricane Matthew imagery revealed a drop in generalization performance (best mean IoU: 46.2%), pointing to domain shift challenges. This key limitation highlights variations in sensor acquisition conditions, land cover, and flood characteristics across events. To address this, future studies could explore domain adaptation techniques such as incorporating multi-temporal data or event-agnostic pretraining on large, diverse datasets to improve robust model training and prediction. Also, fine-tuning with limited labeled samples from target domains or leveraging self-supervised pretraining could enhance generalization to new flood events. Overall, the findings support the effectiveness of multi-modal fusion—especially UAV RGB with UAVSAR VV (5 m) for fine-grained flood mapping and suggest future work in multi-temporal and domain-adaptive modeling.

REFERENCES

- [1] H. Tabari, “Climate change impact on flood and extreme precipitation increases with water availability,” *Sci. Rep.*, vol. 10, no. 1, p. 13768, Aug. 2020.
- [2] M. Zachariah, V. Kotroni, L. Kostas, C. Barnes, J. Kimutai, S. Kew, and I. Pinto, “Interplay of climate change-exacerbated rainfall, exposure and vulnerability led to widespread impacts in the Mediterranean region,” 2023.
- [3] M. S. Tehrani, H. Özener, B. Kalantar, N. Ueda, M. R. Habibi, F. Shabani, V. Saeidi, and F. Shabani, “Application of an ensemble statistical approach in spatial predictions of bushfire probability and risk mapping,” *J. Sensors*, vol. 2021, no. 1, pp. 1–31, Jan. 2021.

- [4] Q. Sun, J. Mann, and M. Skidmore, "The impacts of flooding and business Activity and employment: A spatial perspective on small business," *Water Econ. Policy*, vol. 8, no. 3, Jul. 2022, Art. no. 2140003.
- [5] R. Newman and I. Noy, "The global costs of extreme weather that are attributable to climate change," *Nature Commun.*, vol. 14, no. 1, p. 6103, 2023.
- [6] J. B. Davies, "Economic analysis of the costs of flooding," *Can. Water Resour. J. / Revue Canadienne Des Ressources Hydriques*, vol. 41, nos. 1–2, pp. 204–219, Apr. 2016.
- [7] S. Pralle, "Drawing lines: FEMA and the politics of mapping flood zones," *Climatic Change*, vol. 152, no. 2, pp. 227–237, Jan. 2019.
- [8] C. Fischer and P. Stanchev. (2022). *Flood Hazard and Risk Maps: A Key Instrument for Flood Risk Management*. [Online]. Available: <https://blogs.worldbank.org/water/flood>
- [9] J. F. S. Barrionuevo, H. C. Noblejas, and M. F. M. Rodríguez, "Mapping tools for flood risk rescue and assistance management," *Land*, vol. 13, no. 1, p. 68, Jan. 2024.
- [10] A. Shastry, E. Carter, B. Coltin, R. Sleeter, S. McMichael, and J. Eggleston, "Mapping floods from remote sensing data and quantifying the effects of surface obstruction by clouds and vegetation," *Remote Sens. Environ.*, vol. 291, Jun. 2023, Art. no. 113556.
- [11] R. Yadav, A. Nascetti, and Y. Ban, "Deep attentive fusion network for flood detection on uni-temporal Sentinel-1 data," *Frontiers Remote Sens.*, vol. 3, Dec. 2022, Art. no. 1060144.
- [12] C. Dey, X. Jia, and D. Fraser, "Decision fusion for reliable flood mapping using remote sensing images," in *Proc. Digit. Image Comput., Techn. Appl.*, Dec. 2008, pp. 184–190.
- [13] F. Yamazaki and M. Matsuoka, "Remote sensing technologies in post-disaster damage assessment," *J. Earthq. Tsunami*, vol. 1, no. 3, pp. 193–210, 2007.
- [14] A. D'Addabbo, A. Refice, F. P. Lovergine, and G. Pasquariello, "DAFNE: A MATLAB toolbox for Bayesian multi-source remote sensing and ancillary data fusion, with application to flood mapping," *Comput. Geosci.*, vol. 112, pp. 64–75, Mar. 2018.
- [15] H. Friedrich, B. Tellman, R. Mukherjee, P. Gentine, V. Lakshmi, U. Lall, A. Kruczkiewicz, S. Keene, E. Colon, and Y. Ofek, "Toward mapping flood impacts in urban areas using multi-sensor satellite data fusion," in *Proc. AGU Fall Meeting Abstr.*, 2021, p. 1537.
- [16] D. Munasinghe, R. P. D. M. Frasson, C. H. David, M. Bonnema, G. Schumann, and G. R. Brakenridge, "A multi-sensor approach for increased measurements of floods and their societal impacts from space," *Commun. Earth Environ.*, vol. 4, no. 1, p. 462, Dec. 2023.
- [17] J. Yang, L. El Mendili, Y. Khayer, S. McArdle, and L. H. Beni, "Instance segmentation of LiDAR data with vision transformer model in support of inundation mapping under forest canopy environment," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 48, pp. 203–208, Dec. 2023.
- [18] L. Hashemi-Beni, M. Puthenparampil, and A. Jamali, "A low-cost IoT-based deep learning method of water gauge measurement for flood monitoring," *Geomatics, Natural Hazards Risk*, vol. 15, no. 1, Dec. 2024, Art. no. 2364777.
- [19] W. Tan, N. Qin, Y. Zhang, H. McGrath, M. Fortin, and J. Li, "A rapid high-resolution multi-sensory urban flood mapping framework via DEM upscaling," *Remote Sens. Environ.*, vol. 301, Feb. 2024, Art. no. 113956.
- [20] A. Cerbelaud, G. Blanchet, L. Roupioz, P. Breil, and X. Briottet, "Mapping pluvial flood-induced damages with multi-sensor optical remote sensing: A transferable approach," *Remote Sens.*, vol. 15, no. 9, p. 2361, Apr. 2023.
- [21] X. Du and A. Zare, "Multiresolution multimodal sensor fusion for remote sensing data with label uncertainty," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2755–2769, Apr. 2020.
- [22] A. Salem and L. Hashemi-Beni, "Inundated vegetation mapping using SAR data: A comparison of polarization configurations of UAVSAR L-band and sentinel C-band," *Remote Sens.*, vol. 14, no. 24, p. 6374, Dec. 2022.
- [23] P. Ghamisi, B. Rasti, N. Yokoya, Q. Wang, B. Hofile, L. Bruzzone, F. Bovolo, M. Chi, K. Anders, R. Gloaguen, P. M. Atkinson, and J. A. Benediktsson, "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.
- [24] M. J. Olsen, Z. Chen, T. Hutchinson, and F. Kuester, "Optical techniques for multiscale damage assessment," *Geomatics, Natural Hazards Risk*, vol. 4, no. 1, pp. 49–70, Mar. 2013.
- [25] J. Fayne, J. Bolten, V. Lakshmi, and A. Ahamed, "Optical and physical methods for mapping flooding with satellite imagery," in *Remote Sensing of Hydrological Extremes*. Springer, 2017, pp. 83–103.
- [26] M. Gianinetto, P. Villa, and G. Lechi, "Postflood damage evaluation using Landsat TM and ETM+ data integrated with DEM," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 1, pp. 236–243, Jan. 2006.
- [27] S. A. Ahmadi, A. Mohammadzadeh, N. Yokoya, and A. Ghorbanian, "BD-SKUNet: Selective-kernel UNets for building damage assessment in high-resolution satellite images," *Remote Sens.*, vol. 16, no. 1, p. 182, Dec. 2023.
- [28] F. Foroughnia, V. Macchiarulo, L. Berg, M. DeJong, P. Milillo, K. W. Hudnut, K. Gavin, and G. Giardina, "Quantitative assessment of earthquake-induced building damage at regional scale using LiDAR data," *Int. J. Disaster Risk Reduction*, vol. 106, May 2024, Art. no. 104403.
- [29] M. Rahnemounfar, T. Chowdhury, and R. Murphy, "RescueNet: A high resolution UAV semantic segmentation benchmark dataset for natural disaster damage assessment," 2022, *arXiv:2202.12361*.
- [30] X. Tong, X. Luo, S. Liu, H. Xie, W. Chao, S. Liu, S. Liu, A. N. Makhinov, A. F. Makhinova, and Y. Jiang, "An approach for flood monitoring by the combined use of Landsat 8 optical imagery and COSMO-SkyMed radar imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 136, pp. 144–153, Feb. 2018.
- [31] B. Weintrit, K. Osńska-Skotak, and M. Pilarska, "Feasibility study of flood risk monitoring based on optical satellite data," *Miscellanea Geographica*, vol. 22, no. 3, pp. 172–180, Sep. 2018.
- [32] D. Ehrlich and P. Tenerelli, "Optical satellite imagery for quantifying spatio-temporal dimension of physical exposure in disaster risk assessments," *Natural Hazards*, vol. 68, no. 3, pp. 1271–1289, Sep. 2013.
- [33] A. Beamish, M. K. Reynolds, H. Epstein, G. V. Frost, M. J. Macander, H. Bergstedt, A. Bartsch, S. Kruse, V. Miles, C. M. Tanis, B. Heim, M. Fuchs, S. Chabrilat, I. Shevtsova, M. Verdonen, and J. Wagner, "Recent trends and remaining challenges for optical remote sensing of Arctic Tundra vegetation: A review and outlook," *Remote Sens. Environ.*, vol. 246, Sep. 2020, Art. no. 111872.
- [34] E. Ibrahim, J. Jiang, L. Lema, P. Barnabé, G. Giuliani, P. Lacroix, and E. Pirard, "Cloud and cloud-shadow detection for applications in mapping small-scale mining in Colombia using Sentinel-2 imagery," *Remote Sens.*, vol. 13, no. 4, p. 736, Feb. 2021.
- [35] J. Wang, D. Yang, S. Chen, X. Zhu, S. Wu, M. Bagonovich, Z. Guo, Z. Zhu, and J. Wu, "Automatic cloud and cloud shadow detection in tropical areas for PlanetScope satellite images," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112604.
- [36] T. W. Gillespie, J. Chu, E. Frankenberg, and D. Thomas, "Assessment and prediction of natural hazards from satellite imagery," *Prog. Phys. Geography, Earth Environ.*, vol. 31, no. 5, pp. 459–470, Oct. 2007.
- [37] S. Plank, "Rapid damage assessment by means of multi-temporal SAR—A comprehensive review and outlook to Sentinel-1," *Remote Sens.*, vol. 6, no. 6, pp. 4870–4906, May 2014.
- [38] Y.-N. Du, D.-C. Feng, and G. Wu, "InSAR-based rapid damage assessment of urban building portfolios following the 2023 Turkey earthquake," *Int. J. Disaster Risk Reduction*, vol. 103, Mar. 2024, Art. no. 104317.
- [39] H. Liu, C. Song, Z. Li, Z. Liu, L. Ta, X. Zhang, B. Chen, B. Han, and J. Peng, "A new method for the identification of earthquake-damaged buildings using Sentinel-1 multi-temporal coherence optimized by homogeneous SAR pixels and histogram matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7124–7143, 2024.
- [40] V. Patil, Y. Khadke, A. Joshi, and S. Sawant, "Flood mapping and damage assessment using ensemble model approach," *Sens. Imag.*, vol. 25, no. 1, p. 15, Mar. 2024.
- [41] C. Wang, T. M. Pavelsky, F. Yao, X. Yang, S. Zhang, B. Chapman, C. Song, A. Sebastian, B. G. Frizzelle, E. Frankenberg, and N. Clinton, "Flood extent mapping during hurricane Florence with repeat-pass L-band UAVSAR images," *Water Resour. Res.*, vol. 58, no. 3, 2021, Art. no. e2021WR030606.
- [42] D. Notti, D. Giordan, F. Caló, A. Pepe, F. Zucca, and J. P. Galve, "Potential and limitations of open satellite data for flood mapping," *Remote Sens.*, vol. 10, no. 11, p. 1673, Oct. 2018.

- [43] M. H. Asad, M. M. Asim, M. N. M. Awan, and M. H. Yousaf, "Natural disaster damage assessment using semantic segmentation of UAV imagery," in *Proc. Int. Conf. Robot. Autom. Ind. (ICRAI)*, Mar. 2023, pp. 1–7.
- [44] S. I. Jiménez-Jiménez, W. Ojeda-Bustamante, R. E. Ontiveros-Capurata, and M. D. J. Marcial-Pablo, "Rapid urban flood damage assessment using high resolution remote sensing data and an object-based approach," *Geomatics, Natural Hazards Risk*, vol. 11, no. 1, pp. 906–927, Jan. 2020.
- [45] M. Loli, J. Manousakis, S. A. Mitoulis, and D. Zekkos, "Uav-enabled flood damage assessment and recovery monitoring of bridges following medicane ianos," in *Bridge Safety, Maintenance, Management, Life-Cycle, Resilience and Sustainability*. Boca Raton, FL, USA: CRC Press, 2022, pp. 828–836.
- [46] A. Sharma and U. Verma, "Flood magnitude assessment from UAV aerial videos based on image segmentation and similarity," in *Proc. IEEE Region 10 Conf. (TENCON)*, Dec. 2021, pp. 476–481.
- [47] L. Wouters, H. de Moel, M. C. De Ruiter, A. Couasnon, M. J. C. van den Homberg, and A. Teklesadik, "Improving flood damage assessments in data scarce areas by retrieval of building characteristics through uav image segmentation and machine learning—A case study of the 2019 floods in Southern Malawi," *Natural Hazards Earth Syst. Sci. Discuss.*, vol. 2021, pp. 1–22, Jan. 2021.
- [48] J. Zawadzka, I. Truckell, A. Khouakhi, and M. Rivas Casado, "Detection of flood damage in urban residential areas using object-oriented UAV image analysis coupled with tree-based classifiers," *Remote Sens.*, vol. 13, no. 19, p. 3913, Sep. 2021.
- [49] W. Elmenreich, "Sensor fusion in time-triggered systems," Ph.D. thesis, Technische Universität, Vienna, Austria, 2002.
- [50] C. Pohl and J. L. Van Genderen, "Review article multisensor image fusion in remote sensing: Concepts, methods and applications," *Int. J. Remote Sens.*, vol. 19, no. 5, pp. 823–854, Jan. 1998.
- [51] H. Ghassemlan, "A review of remote sensing image fusion methods," *Inf. Fusion*, vol. 32, pp. 75–89, Nov. 2016.
- [52] N.-B. Chang, K. Bai, S. Imen, C.-F. Chen, and W. Gao, "Multi-sensor satellite image fusion and networking for all-weather environmental monitoring," *IEEE Syst. J.*, vol. 12, no. 2, pp. 1341–1357, Jun. 2018.
- [53] T. G. Yitayew, C. Brekke, and A. P. Doulgeris, "Multisensor data fusion and feature extraction for forestry applications," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 4982–4985.
- [54] Y. Li, S. Martinis, M. Wieland, S. Schlaffer, and R. Natsuaki, "Urban flood mapping using SAR intensity and interferometric coherence via Bayesian network fusion," *Remote Sens.*, vol. 11, no. 19, p. 2231, Sep. 2019.
- [55] D. F. Muñoz, P. Muñoz, H. Moftakhari, and H. Moradkhani, "From local to regional compound flood mapping with deep learning and data fusion techniques," *Sci. Total Environ.*, vol. 782, Aug. 2021, Art. no. 146927.
- [56] M. Fawakherji, J. Blay, M. Anokye, L. Hashemi-Beni, and J. Dorton, "DeepFlood for inundated vegetation high-resolution dataset for accurate flood mapping and segmentation," *Sci. Data*, vol. 12, no. 1, p. 271, Feb. 2025.
- [57] M. Fawakherji and L. Hashemi-Beni, "Flood detection and mapping through multi-resolution sensor fusion: Integrating UAV optical imagery and satellite SAR data," *Geomatics, Natural Hazards Risk*, vol. 16, no. 1, Dec. 2025, Art. no. 2493225.
- [58] Y. Wang, H. Wang, J. Hu, and H. Gui, "Optimization design of miniature air quality monitoring system based on multi-sensor fusion technology," *J. Atmos. Environ. Opt.*, vol. 16, no. 4, p. 349, 2021.
- [59] J. Blay, M. Fawakherji, M. Anokye, and L. H. Beni, "Insights into multi-resolution remote sensing data fusion: Utilizing UAV and SAR imagery for enhanced flood mapping," in *Proc. AGU Fall Meeting Abstr.*, 2024, p. 14.
- [60] F. Wang and X. Feng, "Flood change detection model based on an improved U-Net network and multi-head attention mechanism," *Sci. Rep.*, vol. 15, no. 1, p. 3295, Jan. 2025.
- [61] S. Kundu, V. Lakshmi, and R. Torres, "Estimation of flood inundation and depth during hurricane Florence using Sentinel-1 and UAVSAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [62] R. Liu, J. Ling, and H. Zhang, "SoftFormer: SAR-optical fusion transformer for urban land use and land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 277–293, Dec. 2024.
- [63] H. Wang, H. Wang, and L. Wu, "TGF-Net: Transformer and GIST CNN fusion network for multi-modal remote sensing image classification," *PLoS ONE*, vol. 20, no. 2, Feb. 2025, Art. no. e0316900.
- [64] D. Hong, B. Zhang, H. Li, Y. Li, J. Yao, C. Li, M. Werner, J. Chanutot, A. Zipf, and X. X. Zhu, "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.
- [65] D. Ibañez, J. Xia, N. Yokoya, F. Pla, and R. Fernandez-Beltran, "Inter-sensor high-resolution and multi-temporal image fusion for unsupervised domain adaptation in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
- [66] B. Yu, J. Li, and X. Huang, "STSNNet: A cross-spatial resolution multi-modal remote sensing deep fusion network for high resolution land-cover segmentation," *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102689.
- [67] D. L. Paterson, H. Wright, and P. N. A. Harris, "Health risks of flood disasters," *Clin. Infectious Diseases*, vol. 67, no. 9, pp. 1450–1454, 2018.
- [68] A. Freeman and S. L. Durden, "A three-component scattering model for polarimetric SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 3, pp. 963–973, 2002.
- [69] J.-S. Lee, M. R. Grunes, and G. de Grandi, "Polarimetric SAR speckle filtering and its implication for classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 5, pp. 2363–2373, May 1999.
- [70] M. N. S. Ramya and S. Kumar, "PolInSAR coherence-based decomposition modeling for scattering characterization: A case study in Uttarakhand, India," *Sci. Remote Sens.*, vol. 3, Jun. 2021, Art. no. 100020.
- [71] B. Cheng, R. Girshick, P. Dollár, A. C. Berg, and A. Kirillov, "Boundary IoU: Improving object-centric image segmentation evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15329–15337.
- [72] A. Garcia-Garcia, S. Orts-Escolano, S. Oprea, V. Villena-Martinez, and J. Garcia-Rodriguez, "A review on deep learning techniques applied to semantic segmentation," 2017, *arXiv:1704.06857*.
- [73] A. Abbasi, C. De Sousa, and A. Karami, "Sentinel-1 polarization comparison for flood segmentation using deep learning," *Proceedings*, vol. 87, no. 1, p. 14, 2023.
- [74] H. Wu, H. Song, J. Huang, H. Zhong, R. Zhan, X. Teng, Z. Qiu, M. He, and J. Cao, "Flood detection in dual-polarization SAR images based on multi-scale deeplab model," *Remote Sens.*, vol. 14, no. 20, p. 5181, Oct. 2022.
- [75] J. Kim, S. Li, and W. Zhao, "On flood detection using dual-polarimetric SAR imagery and deep learning approaches," *Remote Sens.*, vol. 17, no. 11, p. 1931, 2025.
- [76] T. Dang, "Comparison of polarization models in SAR for water body extraction and flood mapping," Master's thesis, KTH Royal Inst. of Technol., Stockholm, Sweden, 2024.
- [77] Y. Wang, J. Kim, S. Hensley, and B. Chapman, "Flood extent mapping during hurricane Florence with repeat quad-polarized UAVSAR data," *Water Resour. Res.*, vol. 58, no. 12, 2022, Art. no. e2021WR030606.



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